Aidan Ream

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Overview

Computer Graphics Programmer with experience in both high-level applications and low-level implementation. Focused on a blended approach in Animation Systems programming and implementation.

Projects

Monte Carlo Ray Tracer

- Made a C++ Path Tracer with the following features: Geometric primitive and triangle-mesh support, performance optimization structures, multi-threading, png output, Probability Distribution Functions, Atmospheric Persepective camera effects, and more.
- Tools Used: C++

Compute Shader Based Vulkan Cloth Simluation

- Created a real time simulation environment with the Vulkan graphics library that utilizes a spring based compute shader on a grid of connected points to simulate the movement and flow of cloth using the GPU for performance optimization.
- Tools Used: C++, Vulkan

Skeletal Animation System For Custom Game Engine

- Collaborated on a 3D Roguelike adventure game created by building a custom game engine with OpenGL. My primary contributions were implementing a skeletal animation system, 3D modeling and animations, and creating the debug mode.
- Tools Used: C++, openGL, Blender

Dynamic Dithering Shader

- Created a Shader Algorithm to introduce “dithering” printing effect to a dynamic 3D Scene
- Tools Used: C++, openGL, GLSL, GLFW

Technology Demo for Expressive Technology Studios

- Built an immersive technology demo in Unity combining motion capture, blender animation, spatial audio.
- Tools Used: Unity, Blender, Rokoko studio, Davinci Resolve

Experience

Calpoly, Lab Manager at Expressive Technology Studios — San Luis Obispo, CA — August 2022 – December 2025

- Provided software updates and equipment upkeep for lab equipment over the course of 3.5 years
- Ran 10+ events per year utilizing our spatial audio theater space, assisting visiting speakers with the use of the software

ACM SIGGRAPH, Student Volunteer — Vancouver, BC - August 2025; Los Angeles, CA - July 2026

- Provided stage assistance during four performances in the ’Spatial Storytelling’ Event
- Assisted conference attendees with navigation and scheduling

Skills

Programming Languages: C++, C, Java, Python, GLSL, JavaScript

Graphics Libraries & Game Engines: openGL, Vulkan, Unity, Godot, Houdini, RenderDoc

Visual Design Software: Maya, Blender, Substance Painter, Photoshop, Adobe Illustrator, Krita

Soft skills: Team Management, Group Communication, Agile/Scrum workflows

Education

Calpoly, San Luis Obispo CA — Graduated June 2026

Bachelor of Science in Liberal Arts and Engineering Studies — Cumulative GPA 3.54

Computer Graphics concentration - Coursework focused on computer graphics projects

2D and 3D Animation concentration - Coursework focused on design principles and industry standard workflows/technologies

Computer Science Minor - Additional coursework taken exploring various computer science disciplines